

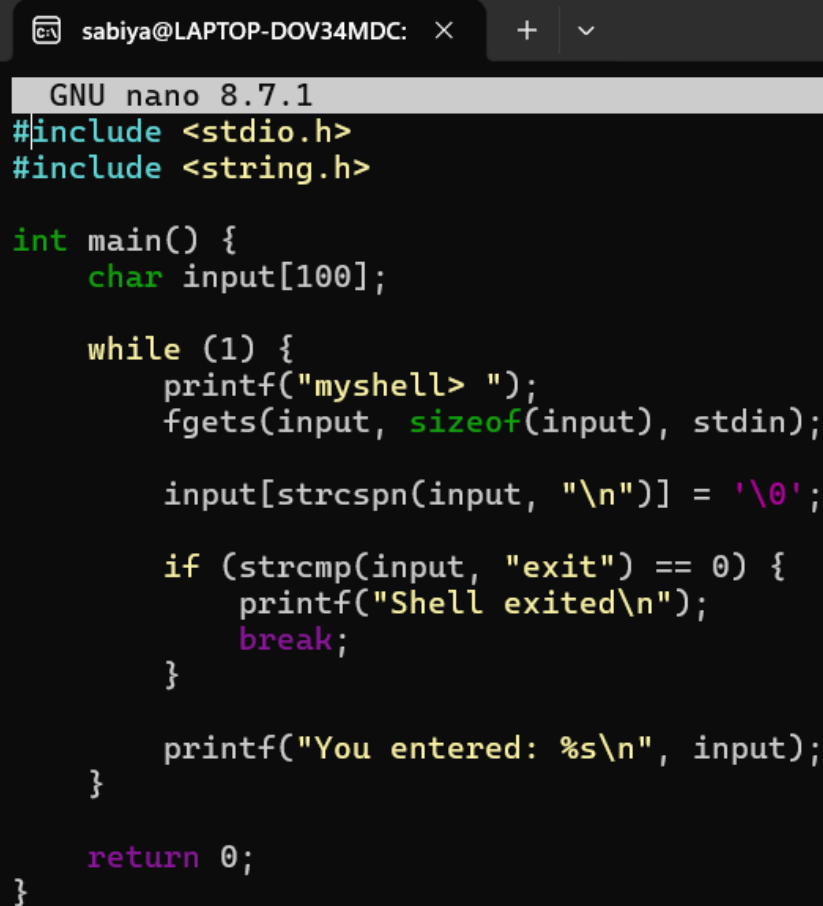
OSSP (Operating Systems And Systems Programming) - 25CS2104E

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1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

1(a) Mainloop code:



```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myshell> ");
        fgets(input, sizeof(input), stdin);

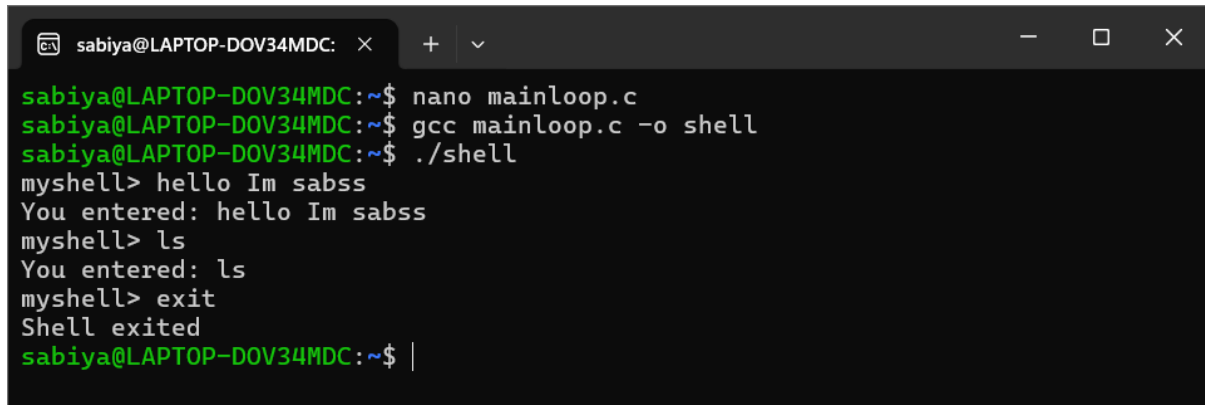
        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            printf("Shell exited\n");
            break;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}
```

OUTPUT SCREENSHOT:



```
sabiya@LAPTOP-DOV34MDC:~$ nano mainloop.c
sabiya@LAPTOP-DOV34MDC:~$ gcc mainloop.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
myshell> hello Im sabss
You entered: hello Im sabss
myshell> ls
You entered: ls
myshell> exit
Shell exited
sabiya@LAPTOP-DOV34MDC:~$ |
```

1(b) Capture Keyboard Input code:



```
GNU nano 3.7.1 keyboard.c
#include <stdio.h>
#include <termios.h>
#include <unistd.h>

int main() {
    struct termios old, new;
    char ch;
    char buffer[100];
    int i = 0;

    tcgetattr(STDIN_FILENO, &old);
    new = old;
    new.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &new);

    printf("Enter command: ");

    while (1) {
        ch = getchar();

        if (ch == '\n') {
            buffer[i] = '\0';
            break;
        }

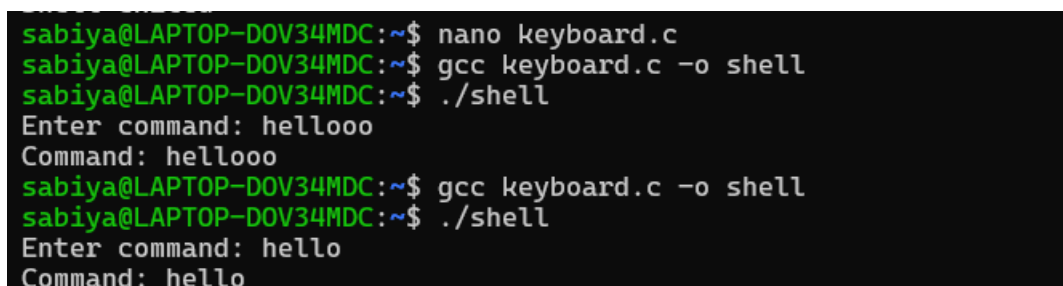
        if (ch == 127) {
            if (i > 0) {
                i--;
                printf("\b \b");
            }
        }

        else if (i < 99) {
            buffer[i++] = ch;
            printf("%c", ch);
        }
    }

    tcsetattr(STDIN_FILENO, TCSANOW, &old);
    printf("\nCommand: %s\n", buffer);

    return 0;
}
```

OUTPUT SCREENSHOT:



```
sabiya@LAPTOP-DOV34MDC:~$ nano keyboard.c
sabiya@LAPTOP-DOV34MDC:~$ gcc keyboard.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Enter command: hellooo
Command: hellooo
sabiya@LAPTOP-DOV34MDC:~$ gcc keyboard.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Enter command: hello
Command: hello
```